

TWO EXACT POSITIVE-WITNESS TESTS FOR ERDŐS–STRAUS DIVISOR BOXES

THE CLANKERS RESEARCH WORKBENCH

ABSTRACT. We retain all multiplicities in a signed divisor box and derive a sharp integer collision threshold for meeting the combined exterior/middle target. It forces a marked witness at $p = 1201$ where the ordinary Cauchy–Schwarz threshold does not. Separately, an exact largest-divisor formula sharpens the cofactor step in an existing angular construction. The bounds are unconditional sufficient criteria with explicit witness inverses, not hypotheses asserted for all primes. Standard finite inequalities, the ES channels, and the previous anchor construction are not claimed as new; historical priority of these particular packaged refinements is unassessed.

Let p be a prime $1 \pmod{4}$, let $p/4 < a < 3p/4$, and put $R = 4a - p$. Every $u \mid a^2$ gives the exterior and middle gates

$$R \mid 4u + 1 \quad (E), \quad R \mid u + a \quad (M).$$

With $g = \gcd(a, u)$, set $h = g^2/u$, $r = u/g$, $s = a/g$. Then $a = hrs$, $u = hr^2$, $\gcd(r, s) = 1$, and the positive ordered witnesses are

$$(hrs, hsk\kappa, phr\kappa), \quad \kappa = (pr + s)/R \quad (E),$$

$$(hrs, phs\lambda, phr\lambda), \quad \lambda = (r + s)/R \quad (M).$$

Integrality follows from the indicated gate. These are the inherited Yamamoto/Type-I–II coordinates [1, 2].

1. THE INTEGER COLLISION THRESHOLD

Write $a = \prod q^{e_q}$ and let $\mu(g)$ count the vectors $\beta \in \prod_q [-e_q, e_q] \cap \mathbb{Z}^{\omega(a)}$ satisfying $\prod q^{\beta_q} = g$ in $G = (\mathbb{Z}/R\mathbb{Z})^\times$. Set

$$N = \prod_q (2e_q + 1), \quad E = \sum_{g \in G} \mu(g)^2, \quad \mathcal{T} = \{-1, -p, -p^{-1}\} \subset G.$$

Coincident targets are counted once. The inversion involution gives $\mu(g) = \mu(g^{-1})$; both channels fail exactly when μ vanishes on $\mathcal{T}$.

Theorem 1. *Let $M = |G| - |\mathcal{T}| > 0$, write $N = Mq + r$ with $0 \leq r < M$, and put $L = Mq^2 + r(2q + 1)$. If $E < L$, one of the two gates has an actual divisor hit. The energy can be computed without discarding any exponent budget by*

$$E = \sum_{\delta_q = -2e_q}^{2e_q} \mathbf{1}_{\prod q^{\delta_q} = 1 \pmod{R}} \prod_q (2e_q + 1 - |\delta_q|).$$

Proof. If there is no hit, N occurrences occupy M allowed bins. Whenever two counts differ by two or more, moving one occurrence from the larger to the smaller strictly lowers their sum of squares. Hence the minimum has r counts $q + 1$ and $M - r$ counts q , proving $E \geq L$. The displayed energy formula counts ordered pairs of exponent vectors of equal image, grouped by their difference. In coordinate q a difference δ_q occurs in $2e_q + 1 - |\delta_q|$ ordered pairs.

For the inverse, take an actual vector β in a target fibre. If its image is -1 , use M ; if it is $-p^{-1}$, use E ; if it is only $-p$, first negate every exponent and use E . In all cases $u = \prod q^{e_q + \beta_q}$ has exponents in $[0, 2e_q]$. The gate and the ordered denominator formulas then hold exactly. $\square$

There is a further exact parity improvement. If $1 \in \mathcal{T}$, the zero vector is a hit. Otherwise let s count the allowed nonidentity involutions and r the allowed unordered inverse pairs. A no-hit histogram has identity count $1 + 2x_0$, involution counts $2x_i$, and pair counts y_j, y_j . Consequently its exact relaxed energy minimum is

$$\min_{x_0 + \dots + x_s + y_1 + \dots + y_r = (N-1)/2} \left((1 + 2x_0)^2 + 4 \sum_{i=1}^s x_i^2 + 2 \sum_{j=1}^r y_j^2 \right).$$

It is computed by taking the required number of least marginal costs from 8, 16, 24, ... (identity), 4, 12, 20, ... (each involution), and 2, 6, 10, ... (each inverse pair). The increasing marginal sequences make every selected prefix feasible; an exchange proves optimality.

At $p = 1201, a = 306, R = 23$, the actual box is $[-1, 1] \times [-2, 2] \times [-1, 1]$ on primes 2, 3, 17. The energy is 107, the target set is $\{9, 18, 22\}$, and $N = 45, M = 19$. Ordinary Cauchy-Schwarz does not force a hit: $19 \cdot 107 = 2033 > 45^2 = 2025$. But the integer threshold is

$$12 \cdot 2^2 + 7 \cdot 3^2 = 111 > 107.$$

The inverse gives $u = 17, E$, and (306, 16218, 1082101). This is a better sufficient criterion, not a claim of new numerical coverage. The full multiplicity table is in the certificate.

2. A SHARPER ANGULAR COFACTOR STEP

Let $H \leq (\mathbb{Z}/R\mathbb{Z})^\times$ have index two, with $-1 \notin H$. Choose an actual divisor $A \mid a$ whose prime factors lie in H and whose pairs $nd \mid A$, $\gcd(n, d) = 1$, represent every $v = n/d \in H$. Define the attained rational costs

$$c_A(v) = \min_{n/d=v \ (R)} d/n, \quad B_A = \max_{v \in H} c_A(v), \quad b = a/A.$$

This hypothesis concerns a finite, explicitly checked anchor; it is not asserted for an arbitrary shell or unknown prime. It is the exact anchor notion from the preceding workbench angular construction [3].

Theorem 2. *The greatest available divisor of b outside H is*

$$T_H(b) = b \quad (b \notin H), \quad T_H(b) = b/q_* \quad (b \in H \text{ with an outside prime factor}),$$

where q_* is the least such prime; it is zero when no outside factor exists. If $T_H(b) > 8B_A$, the shell has a middle witness with $0 < r/s < 1/8$. The exact anchor-wise criterion is $\max_{t \mid b, t \notin H} t/c_A(-t^{-1}) > 8$.

Proof. When $b \in H$ and $t \notin H$, the complement b/t is outside and must contain an outside prime factor. Hence $b/t \geq q_*$. Equality is attained by $t = b/q_*$. The other cases are immediate. This uses one actual prime occurrence even when q_* divides b repeatedly.

For any chosen outside t , take a minimizing pair (n, d) at $-t^{-1}$. Then $R \mid d + tn$ and $dtn \mid a$. Put

$$g = \gcd(d, tn), \quad r = d/g, \quad s = tn/g, \quad h = ag^2/(dtn), \quad u = ad/(tn), \quad \lambda = (d + tn)/(gR).$$

All are positive integers, g is a unit modulo R , $a = hrs$, $u = hr^2$, and $\gcd(r, s) = 1$. The middle gate holds and $r/s = c_A(-t^{-1})/t$. The largest-divisor sufficient condition and exact criterion follow. For a target with the anchor retained, all inverse pairs satisfy $t = ds/(nr) \mid b$; then $r \mid d$, $g = d/r$, and the original packet is recovered. Thus the finite fibres are explicit. $\square$

The source Koide lower boundary exceeds $1/8$, as is already certified by $\sqrt{3} > 1.732, \sqrt{2} < 1.415$ and $317/2415 > 1/8$. The coarse earlier threshold $b > 64B_A^2$ is replaced, when $b \in H$, by the factor-sensitive bound $b > 8B_A q_*$.

For $R = 19, A = 35$, the nine residue costs (residue, cost) are

v	1	4	5	6	7	9	11	16	17
$c_A(v)$	1	5	1/5	35	1/7	5/7	7	1/35	7/5

These values follow by listing the nine coprime pairs $nd \mid 35$. At the prime $p = 558721$, $a = 139685 = 35 \cdot 3991$, $b = 13 \cdot 307 \in H$, $q_* = 13$, and $307 > 280$ but $3991 < 78400$. The minimizing pair $(n, d) = (1, 35)$ yields

$$(R, u, h, r, s, \lambda) = (19, 15925, 13, 35, 307, 18),$$

with denominators $(139685, 40137399198, 4575924990)$. No claim is made that the resulting prime family was previously uncovered.

Scope. Both theorems construct actual witnesses when their proved inequalities hold. A counterexample to ES would have to violate every sufficient test in every full shell. No theorem that one of these tests always succeeds is assumed. The standard-library verifier recalculates the energies, their full fibres, the symmetric minima, all anchor pairs, and the explicit witness returns.

REFERENCES

- [1] Yamamoto, K. (1965). On the Diophantine equation $4/n = 1/x + 1/y + 1/z$. *Mem. Fac. Sci. Kyushu Univ. A*, 19(1), 37–47. Lemma 2, equation (4), p. 38.
- [2] Elsholtz, C., & Tao, T. (2013). Counting the number of solutions to the Erdős–Straus equation on unit fractions. *J. Aust. Math. Soc.*, 94(1), 50–105. §2.
- [3] The Clankers research workbench (2026). *Pointwise selector continuation*, section “Weighted angular forcing”. The supplied predecessor `es_pointwise_continuation_bundle.zip` retains the original finite anchor and its complete middle-state inverse. The largest-cofactor step and integer collision refinement are the present continuation. No historical-priority claim is made.